Complete positivity range of the qubit depolarizing map

Alex

October 14, 2026

Setting

Sources. The proof below is self-contained. The only result from the user's papers is Choi's criterion, and it appears only in the optional cross-check: [1, unnumbered statement before Lemma 67, p. 25 (PDF page)]. Other standard facts are named where they are used.

Hypotheses.

H1. $\mathcal{H} = \mathbb{C}^2$ (finite-dimensional). $p \in \mathbb{R}$.

H2. Φ_p is defined by the formula in Theorem 1 on all of $M_2(\mathbb{C})$. It is linear.

H3. Completely positive (CP) means: for every $n \geq 1$, $\Phi_p \otimes \mathrm{id}_{M_n}$ maps positive semidefinite (PSD) elements of $M_2 \otimes M_n$ to PSD elements.

Conventions.

- Physics inner product (linear in the second slot). $\{|0\rangle, |1\rangle\}$ is the computational basis and $E_{ij} = |i\rangle\langle j|$.
- Choi matrix (unnormalized, with Φ on the **second** factor): $J(\Phi) = \sum_{i,j} E_{ij} \otimes \Phi(E_{ij})$. This is what `tools/qmath.py:choi` computes.
- $|\Omega\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, so $\sum_{i,j} E_{ij} \otimes E_{ij} = 2\,|\Omega\rangle\langle\Omega|$.
- Paulis: $\sigma_0 = \mathbb{1}$, $\sigma_1 = \sigma_x$, $\sigma_2 = \sigma_y$, $\sigma_3 = \sigma_z$. The identity matrix is $\mathbb{1}$ ($\mathbb{1}_2$, $\mathbb{1}_4$, $\mathbb{1}_n$ when the size matters).

Theorem 1. *For real p let $\Phi_p : M_2(\mathbb{C}) \to M_2(\mathbb{C})$, $\Phi_p(X) = (1-p)X + p\,\mathrm{Tr}(X)\,\mathbb{1}/2$.*

(a) $\mathrm{Tr}\,\Phi_p(X) = \mathrm{Tr}\,X$ *for all $X \in M_2$ and all real p.*

(b) Φ_p *is completely positive iff* $0 \leq p \leq 4/3$.

Strategy. Part (a) is a one-line computation. For (b), the proof is self-contained and does not need Choi's theorem.

- **Necessity:** apply $\Phi_p \otimes \mathrm{id}_2$ to the PSD matrix $2\,|\Omega\rangle\langle\Omega|$. The result is $2(1-p)\,|\Omega\rangle\langle\Omega| + \frac{p}{2}\mathbb{1}_4$, with eigenvalues $2 - \frac{3p}{2}$ (once) and $\frac{p}{2}$ (three times). It fails to be PSD exactly when $p < 0$ or $p > 4/3$.

- **Sufficiency:** for $0 \le p \le 4/3$, write Φ_p in Kraus form, $\Phi_p(X) = \sum_\mu K_\mu X K_\mu^\dagger$ with $K_0 = \sqrt{1 - 3p/4}\,\mathbb{1}$ and $K_k = \sqrt{p/4}\,\sigma_k$. Any such map is CP.

Choi's theorem gives the same result from the spectrum of $J(\Phi_p)$; it is noted as a cross-check.

Proof. 1. **(Part (a).)** For any $X \in M_2$:

$$\operatorname{Tr}\Phi_p(X) = (1-p)\operatorname{Tr}X + p\operatorname{Tr}X \cdot \frac{\operatorname{Tr}\mathbb{1}_2}{2} = (1-p)\operatorname{Tr}X + p\operatorname{Tr}X = \operatorname{Tr}X. \tag{1}$$

This uses linearity of the trace and $\operatorname{Tr}\mathbb{1}_2 = 2$. It holds for every real p, and indeed for every complex p. [Own reasoning], [Computed] (C3).

2. **(Choi matrix.)** By linearity, $\Phi_p(E_{ij}) = (1-p)E_{ij} + \frac{p}{2}\delta_{ij}\mathbb{1}$, because $\operatorname{Tr}E_{ij} = \delta_{ij}$. So

$$J(\Phi_p) = (1-p)\sum_{i,j} E_{ij} \otimes E_{ij} + \frac{p}{2}\sum_i E_{ii} \otimes \mathbb{1} = 2(1-p)\,|\Omega\rangle\langle\Omega| + \frac{p}{2}\mathbb{1}_4, \tag{2}$$

using $\sum_i E_{ii} = \mathbb{1}_2$. [Own reasoning], [Computed] (C1).

3. **(Spectrum.)** $2\,|\Omega\rangle\langle\Omega|$ is $2\times$ a rank-one projector. So $J(\Phi_p)$ has:

   - eigenvalue $2(1-p) + \frac{p}{2} = 2 - \frac{3p}{2}$ on $|\Omega\rangle$;
   - eigenvalue $\frac{p}{2}$ on the 3-dimensional subspace $|\Omega\rangle^\perp$ (for example $|01\rangle$, $|10\rangle$, $(|00\rangle - |11\rangle)/\sqrt{2}$).

   Hence $J(\Phi_p) \ge 0 \iff p \ge 0$ and $p \le 4/3$. [Own reasoning] (spectral theorem for a Hermitian matrix that is a combination of $\mathbb{1}$ and a projector), [Computed] (C2).

4. **(The test matrix under $\Phi_p \otimes \mathrm{id}$.)** Let $P = \sum_{i,j} E_{ij} \otimes E_{ij} = 2\,|\Omega\rangle\langle\Omega| \in M_2 \otimes M_2$. It is PSD because it is $2\times$ a projector. Then

$$(\Phi_p \otimes \mathrm{id}_2)(P) = \sum_{i,j} \Phi_p(E_{ij}) \otimes E_{ij} = 2(1-p)\,|\Omega\rangle\langle\Omega| + \frac{p}{2}\mathbb{1}_4 = J(\Phi_p). \tag{3}$$

   The middle expression is the same computation as in Step 2 with the tensor factors swapped. The swap leaves both $|\Omega\rangle\langle\Omega|$ and $\mathbb{1}_4$ unchanged. So the hypothesis H3 convention (Φ on the first factor) and the Choi convention (Φ on the second factor) give the identical matrix here. [Own reasoning], [Computed] (C8).

5. **(Necessity.)** Suppose Φ_p is CP. Apply H3 with $n = 2$ to the PSD matrix P from Step 4: then $(\Phi_p \otimes \mathrm{id}_2)(P) = J(\Phi_p) \ge 0$. By Step 3, $0 \le p \le 4/3$.

   The contrapositive gives explicit witnesses:

   - If $p < 0$: $\langle 01|J(\Phi_p)|01\rangle = p/2 < 0$.
   - If $p > 4/3$: $\langle\Omega|J(\Phi_p)|\Omega\rangle = 2 - 3p/2 < 0$.

   In both cases $\Phi_p \otimes \mathrm{id}_2$ is not even positive. [Own reasoning], [Computed] (C10, C11).

6. **(Pauli twirl identity.)** For every $X \in M_2$:

$$\sum_{\mu=0}^{3} \sigma_\mu X \sigma_\mu = 2\operatorname{Tr}(X)\,\mathbb{1}. \tag{4}$$

   Proof. Both sides are linear in X, and $\{\sigma_\nu\}$ is a basis of M_2, so it suffices to check $X = \sigma_\nu$.

- For $\nu = 0$: the left side is $4\mathbb{1}$, and $2\,\mathrm{Tr}(\mathbb{1})\mathbb{1} = 4\mathbb{1}$.
- For $\nu = k \in \{1,2,3\}$: $\sigma_\mu \sigma_k \sigma_\mu = \sigma_k$ if σ_μ commutes with σ_k ($\mu \in \{0, k\}$). It equals $-\sigma_k$ if they anticommute (the other two μ). This uses $\sigma_\mu^2 = \mathbb{1}$ and the fact that distinct non-identity Paulis anticommute. So the sum is $(1+1-1-1)\sigma_k = 0 = 2\,\mathrm{Tr}(\sigma_k)\mathbb{1}$. □

This is a standard result, not from your papers (e.g. Nielsen–Chuang, the depolarizing-channel section). [Own reasoning], [Computed] (C4).

7. **(Kraus form.)** By Step 6, $p\,\mathrm{Tr}(X)\mathbb{1}/2 = \frac{p}{4}\left(X + \sum_{k=1}^{3} \sigma_k X \sigma_k\right)$. Therefore

$$\Phi_p(X) = \left(1 - \tfrac{3p}{4}\right)X + \tfrac{p}{4}\sum_{k=1}^{3} \sigma_k X \sigma_k \qquad \text{(all real } p\text{)}. \tag{5}$$

For $0 \le p \le 4/3$ both coefficients are ≥ 0, so we can set $K_0 = \sqrt{1 - 3p/4}\,\mathbb{1}$ and $K_k = \sqrt{p/4}\,\sigma_k$ ($k = 1,2,3$). Then:

- $\Phi_p(X) = \sum_{\mu=0}^{3} K_\mu X K_\mu^\dagger$, since each σ_k is Hermitian;
- $\sum_\mu K_\mu^\dagger K_\mu = (1 - \frac{3p}{4} + \frac{3p}{4})\mathbb{1} = \mathbb{1}$. This is consistent with Step 1.

[Own reasoning], [Computed] (C5, C6, C7).

8. **(Sufficiency.)** Let $0 \le p \le 4/3$, $n \ge 1$, and $Y \in M_2 \otimes M_n$ with $Y \ge 0$. By Step 7 and linearity:

$$(\Phi_p \otimes \mathrm{id}_n)(Y) = \sum_\mu (K_\mu \otimes \mathbb{1}_n)\, Y\, (K_\mu \otimes \mathbb{1}_n)^\dagger. \tag{6}$$

To check this identity: both sides are linear in Y, and product elements $A \otimes B$ span $M_2 \otimes M_n$, so it suffices to check it on them. On $A \otimes B$ the right side is $\left(\sum_\mu K_\mu A K_\mu^\dagger\right) \otimes B = \Phi_p(A) \otimes B$ by Step 7, which is the left side. (Also checked exactly for a general symbolic $Y \in M_2 \otimes M_2$ by the referee: [Computed], `scratch/ref_depolarizing_cp_check.py`, test 3.) Each term $(K \otimes \mathbb{1})Y(K \otimes \mathbb{1})^\dagger$ is PSD: $\langle v|(K \otimes \mathbb{1})Y(K \otimes \mathbb{1})^\dagger|v\rangle = \langle w|Y|w\rangle \ge 0$ with $|w\rangle = (K \otimes \mathbb{1})^\dagger |v\rangle$. A sum of PSD matrices is PSD. So Φ_p is CP. [Own reasoning], [Computed] (C9, a random numerical spot check with $n = 3$).

9. **(Conclusion.)** Steps 5 and 8 together give: Φ_p is CP $\iff$ $0 \le p \le 4/3$. Together with Step 1, Φ_p is a CPTP map (a quantum channel) exactly for $p \in [0, 4/3]$.

□

Cross-check via Choi's theorem. For n-qubit maps, [1, unnumbered statement before Lemma 67, p. 25 (PDF page)] states that the map $\rho \mapsto 2^n\,\mathrm{Tr}_A[J\,(\rho_A^T \otimes \mathbb{1}_{A'})]$ is completely positive iff $J \ge 0$. [Paper] That paper uses the normalized Choi state $J_\Lambda = (\mathrm{id}_A \otimes \Lambda)(|\Phi\rangle\langle\Phi|)$. For $n = 1$ this equals $J(\Phi_p)/2$, and a positive rescaling does not affect $J \ge 0$. [Own reasoning], [Computed] (`scratch/ref_depolarizing_cp_check.py`, test 1). To apply the criterion we also need $\Phi_p(\rho) = 2\,\mathrm{Tr}_A[J_{\Phi_p}(\rho^T \otimes \mathbb{1})]$. The paper states this reconstruction (its eq. (83)) only for quantum channels, so quoting it for $p \notin [0, 4/3]$ would be circular. In fact it holds for every linear map: [Own reasoning], [Computed] exactly for symbolic real p and general ρ (same script, test 2). The paper also phrases the criterion for maps on states; extending it to all of M_2 by linearity is implicit. The general statement (a linear map $M_d \to M_m$ is CP iff $J(\Phi) \ge 0$) is Choi (1975); see also Watrous, *TQI*, Ch. 2. Its hypotheses hold here: finite dimension and a linear map (H1, H2). Combined with Step 3, it gives (b) at once. The argument above does not depend on it.

Remarks.

- Boundary cases:
  - $p = 0$ is the identity channel; J has rank 1 (eigenvalues $2, 0, 0, 0$).
  - $p = 1$ is the completely depolarizing channel; $J = \mathbb{1}_4/2$.
  - $p = 4/3$ gives $\Phi(X) = \frac{1}{3}\sum_k \sigma_k X \sigma_k$; J has rank 3 (eigenvalues $0, \frac{2}{3}, \frac{2}{3}, \frac{2}{3}$) and $K_0 = 0$.
- For $1 < p \leq 4/3$ the channel is still CP, even though the "weight" $1 - p$ on X is negative.
- Positivity alone (not complete positivity) holds on the larger range $0 \leq p \leq 2$: for a pure state $\Phi_p(|\psi\rangle\langle\psi|)$ has eigenvalues $1 - p/2$ (on $|\psi\rangle$) and $p/2$ (on $|\psi\rangle^\perp$), and every PSD matrix is a nonnegative combination of pure states. So for $4/3 < p \leq 2$ the map Φ_p is positive but not CP. [Own reasoning] (not needed for the claim), [Computed] (C12: the Bloch-vector scaling $r \mapsto (1 - p)r$ checked exactly; exact counterexamples at $p = -1/100$ and $p = 201/100$).

Computational checks

Independent scripts by the `quantum-computer` agent: `scratch/q_depolarizing-cp_*.py` (shared definitions in `scratch/q_depolarizing-cp_common.py`; the prover's own script `scratch/depolarizing_cp.py` was not reused). Notation: $J_p = J(\Phi_p)$; matrices are 2×2 or 4×4 over $\mathbb{C}$.

ID	Claim	Method	Verdict
C0	PSD range of J_p (i); main statement on a grid (ii)	(i) exact `solveset` on the eigenvalues of J_p, plus factored characteristic polynomial; (ii) 256 p values: $[-0.5, 2]$ in steps of 0.01, plus probes at $4/3$, $4/3 \pm 10^{-3}$, $\pm 10^{-3}$; applying $\Phi_p \otimes \mathrm{id}_n$ ($n = 2, 3$) directly	(i) Proved symbolically that $J_p \geq 0 \iff p \in [0, 4/3]$; (ii) Supported: negative eigenvalues appear at exactly the grid points outside $[0, 4/3]$ (innermost ones -0.001 and 1.33433)
C1	Closed form of J_p	exact	Proved symbolically
C2	Spectrum of J_p	exact eigenvalues and eigenvectors	Proved symbolically
C3	Trace preservation	exact, general X	Proved symbolically
C4	Pauli twirl identity	exact	Proved symbolically
C5, C6	Kraus form and $\sum K^\dagger K = \mathbb{1}$	exact, two parametrizations covering $[0, 4/3]$, plus 5 rational values	Proved symbolically (a third, t-parametrization in the script is inconclusive because SymPy can't use $t \leq 1$; not relied on)
C7	`qmath.choi` $= J_p$	exact at $p \in \{0, 1/7, 1/2, 1, 4/3\}$	Exact at 5 rational points
C8	Swapped-factor convention	exact	Proved symbolically
C9	$\Phi_p \otimes \mathrm{id}_3$ preserves PSD	500 random ρ per p, ranks 1/2/3/6; worst $\lambda_{\min} \approx -4 \times 10^{-16}$	Supported (numerical, not a proof)
C10	Boundary spectra	exact rationals	Proved symbolically
C11	Witnesses just outside the range	exact rationals	Proved symbolically
C12	Positive iff $0 \leq p \leq 2$	exact Bloch scaling, `solveset`, exact witnesses	Proved symbolically (the step from states to all PSD matrices is own reasoning)

Scope: everything is for M_2. The CP check for general n rests on the Kraus argument in Step 8, which is exact via C5/C6. C0(ii) and C9 are numerical evidence only.

Open doubts

- None on the main mathematics: every step is elementary, every concrete identity except the numerical checks C0(ii) and C9 was proved symbolically, and the referee found no errors.
- The cross-check citation is to an unnumbered statement in [1, p. 25 (PDF page)]. That paper states it for its own normalized Choi state; the $n = 1$ rescaling is own reasoning.
- Referee (`proof-checker`): **Correct, with minor gaps**. All three gaps are now closed: the Step 8 spanning argument, the circularity of eq. (83) in the cross-check, and the table wording. The note states that C0(i) gives the PSD range of J_p; getting from there to CP uses Steps 5/8 (or Choi).

References

[1] Kaifeng Bu, Weichen Gu, and Arthur Jaffe, *Stabilizer Testing and Magic Entropy via Quantum Fourier Analysis*, arXiv:2306.09292v2 [quant-ph], 11 Aug 2025.